

Disproof of conjecture 00000000153

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Verdict: FALSE

We disprove conjecture 00000000153 as stated. The refutation is unconditional and elementary: the finite discrete system $X = \mathbb{Z}/4 = \{0, 1, 2, 3\}$ with $Tx = x + 1$ is minimal, yet for $x = 0$ the prime-index orbit $\{T^{p_n}x : n \geq 1\}$ is $\{1, 2, 3\}$, whose closure in the discrete topology is itself and is therefore not all of X . In fact the conclusion fails for *every* $x \in X$.

1 The conjecture

Quoted verbatim from `conjectures/00000000153.md`:

English. Conjecture: Minimal transitivity of prime orbits: in any minimal topological dynamical system (X, T) , for every $x \in X$ the orbit closure $\{T^{p_n}x : n \geq 1\}$ equals X .

The Chinese statement filed with the conjecture is reproduced verbatim in `README.md` (together with this English text). It is not typeset here, so that `main.tex` has no CJK font requirement and compiles with a plain `tectonic main.tex`. The two formulations are translations of one another, and both leave the quantifier over x and the class of systems unrestricted; the refutation therefore applies to either reading.

The statement leaves the following definitions implicit; we fix the standard ones, so that the refutation cannot be dismissed as a misreading.

Definition 1 (Topological dynamical system). A topological dynamical system is a topological space X together with a continuous map $T : X \rightarrow X$. (In the standard setting of topological dynamics one usually assumes X compact Hausdorff; the witness below is compact Hausdorff, so it satisfies that convention.)

Definition 2 (Minimal system). (X, T) is *minimal* if every orbit is dense in X ; equivalently, the only closed T -invariant subset of X is X itself. In a minimal system the full orbit $\{T^n x : n \geq 0\}$ is dense for every $x \in X$.

Definition 3 (Orbit closure and the prime sequence). $\overline{A}$ denotes the closure of $A \subseteq X$. The sequence $(p_n)_{n \geq 1} = (2, 3, 5, 7, 11, \dots)$ is the sequence of primes in increasing order, so $p_1 = 2$ and p_n is the n -th prime. The *prime-index orbit* of x is the set

$$P(x) = \{T^{p_n}x : n \geq 1\} = \{T^p x : p \text{ prime}\},$$

the two descriptions agreeing because $n \mapsto p_n$ is a bijection onto the set of primes. The conjecture asserts $P(x) = X$ for every x .

2 The witness: $X = \mathbb{Z}/4$ with the shift

Definition 4 (The system). Let $X = \mathbb{Z}/4 = \{0, 1, 2, 3\}$ with the discrete topology, and let

$$T : X \rightarrow X, \quad Tx = x + 1 \pmod{4}.$$

X is finite, hence compact, and discrete, hence Hausdorff and metric: it is a compact metric space. T is continuous (every map on a discrete space is), so (X, T) is a topological dynamical system in the strictest sense.

The dynamics is a single cycle:

$$0 \xrightarrow{T} 1 \xrightarrow{T} 2 \xrightarrow{T} 3 \xrightarrow{T} 0.$$

Lemma 5 (T is a 4-cycle, hence (X, T) is minimal). *For every $x \in X$ the full orbit $\{T^n x : n \geq 0\}$ equals X . Consequently (X, T) is minimal.*

Proof. Iterating, $T^n x = x + n \pmod{4}$ for all $n \geq 0$. For fixed x , as n runs over $\{0, 1, 2, 3\}$ the values $x + n \pmod{4}$ run over all four residues $x, x + 1, x + 2, x + 3$, i.e. over X ; and $T^4 x = x$. Hence $\{T^n x : n \geq 0\} = X$. Every orbit is therefore dense (indeed equal to X), which is minimality. $\square$

Remark 6. The full orbit of every point is *exactly* X , not merely dense. So the alternative reading of the conjecture that replaces $P(x)$ by the full orbit $\{T^{p_n} x\}$ with “ $p_n = n$ ” is trivially true here; but that is a different, degenerate statement (see Section 5).

3 No prime is divisible by 4

Lemma 7 ($4 \nmid p$ for every prime p). *If p is prime, then $4 \nmid p$.*

Proof. Suppose $4 \mid p$. Then $p = 4k$ for some $k \in \mathbb{Z}$, so p is even, i.e. $2 \mid p$. Since p is prime and $2 \mid p$, the divisor 2 of p must satisfy $2 = 1$ or $2 = p$; the first is false, so $p = 2$. But then $4 \mid 2$ would be required, which is false. Hence $4 \nmid p$. $\square$

Equivalently: the only even prime is 2, and $4 \nmid 2$. Therefore for every prime p ,

$$p \bmod 4 \in \{1, 2, 3\},$$

and the value 0 is never attained. The value set is exactly $\{1, 2, 3\}$, witnessed by

$$2 \bmod 4 = 2, \quad 3 \bmod 4 = 3, \quad 5 \bmod 4 = 1.$$

4 The prime-index orbit omits its base point

Theorem 8 (Prime orbit of x). *For every $x \in X$,*

$$P(x) = \{T^p x : p \text{ prime}\} = X \setminus \{x\}.$$

In particular $P(0) = \{1, 2, 3\}$, and $\overline{P(0)} = \{1, 2, 3\} \neq X$ in the discrete topology.

Proof. Since $T^n x = x + n \pmod{4}$, for a prime p we have

$$T^p x = x + p \pmod{4} = x + (p \bmod 4) \pmod{4}.$$

By Lemma 7, $p \bmod 4 \in \{1, 2, 3\}$, so $T^p x \neq x \pmod{4}$: the base point x is never in $P(x)$.

Conversely, fix $y \in X$ with $y \neq x$, and let $d = y - x \bmod 4 \in \{1, 2, 3\}$. Choose a prime $p \equiv d \pmod{4}$:

$$d = 1 \Rightarrow p = 5, \quad d = 2 \Rightarrow p = 2, \quad d = 3 \Rightarrow p = 3.$$

Then $T^p x = x + d = y \pmod{4}$, so $y \in P(x)$. Hence $P(x) = X \setminus \{x\}$.

For $x = 0$ this is $P(0) = \{1, 2, 3\}$. Because X is discrete, every subset is closed, so $\overline{P(0)} = P(0) = \{1, 2, 3\}$, which omits 0 and is a proper subset of $X = \{0, 1, 2, 3\}$. $\square$

Corollary 9 (Refutation). *(X, T) is a minimal topological dynamical system, but the conjecture's conclusion*

$$\forall x \in X, \quad \overline{\{T^{p_n} x : n \geq 1\}} = X$$

is false. Indeed, for every $x \in X$,

$$\overline{P(x)} = X \setminus \{x\} \subsetneq X,$$

so the conclusion fails at every single point x ; a fortiori it fails at the point $x = 0$ exhibited in the conjecture's own quantifier.

Proof. Minimality is Lemma 5. The closure computation is Theorem 8 together with discreteness. A proper subset of X is not equal to X , so the asserted equality fails for each x . $\square$

5 Caveat: scope of the refutation

This caveat is part of the result and should be read together with it.

1. **The witness has isolated points, and is finite.** $X = \mathbb{Z}/4$ is a finite discrete space, so every point is isolated. Some authors in topological dynamics study only minimal systems on compact metric spaces *without isolated points* (or simply *infinite* spaces). $X = \mathbb{Z}/4$ falls outside that additional, *unstated* hypothesis. If one adds such a hypothesis — “ X is a compact metric space with no isolated points”, or just “ X is infinite” — then the conjecture as modified is not touched by this counterexample, and its status remains open here.
2. **But the file states no such hypothesis.** The statement in `conjectures/00000000153.md` says “in any minimal topological dynamical system (X, T) ”, with no cardinality or perfectness restriction, and the quantification is explicitly “for every $x \in X$ ”. Under the literal statement, $(\mathbb{Z}/4, +1)$ is an admissible minimal system and $x = 0$ is an admissible point, so the literal statement is refuted. The finite case is not a technicality outside the intended domain unless the domain is restricted in writing; it is a minimal topological dynamical system in the standard sense.
3. **If instead “orbit closure” is read as the closure of the full orbit, the conjecture collapses rather than being rescued.** In a minimal system the full orbit $\{T^n x : n \geq 0\}$ is dense by definition, so $\overline{\{T^n x : n \geq 0\}} = X$ holds for every x *by definition* of minimality. Reading the ambiguous notation $\{T^{p_n} x : n \geq 1\}$ as the full orbit (i.e. $p_n = n$) makes the

assertion a tautology for every minimal system. That is a collapse of the statement to its definition, not a substantive theorem and not a repair of the prime-index claim: the file’s notation p_n is the standard one for the n -th prime, and the conjecture is titled “prime orbits”.

In short: the literal conjecture is false, the refutation is elementary and rigorous, and the only way to avoid it is to add a hypothesis (infiniteness / no isolated points) that the file does not state. The sharpness note is that under that added hypothesis the witness is excluded and the conjecture is *untouched* — which is precisely why the statement as written is defective.

Remark 10 (On the role of $p_1 = 2$). The prime sequence starts at $p_1 = 2$, and $p = 2$ is included in $P(x)$. This does not change the conclusion: $2 \bmod 4 = 2 \neq 0$, so even the even prime contributes a nonzero residue and $P(x)$ still omits x .

6 Formalisation

The refutation is formalised in the Lean 4 project under `lean4/` (core Lean only — no Mathlib, no `sorry`). The system is $X = \text{Fin } 4$, the shift is $T x = x + 1$, the iterate is `iter`, and primality is defined directly because core Lean has no `Nat.Prime`. The main formal statements are

```
theorem minimal (x y : Fin 4) : FullOrbit x y
theorem four_not_dvd_of_isPrime (hp : IsPrime p) : not (4 | p)
theorem primeOrbitAt_not_self (x : Fin 4) : not (PrimeOrbitAt x x)
theorem primeOrbitClosure_not_univ (x : Fin 4) :
  not (forall y : Fin 4, PrimeOrbitClosure x y)
theorem conjecture_00000000153_false :
  IsMinimal and forall x : Fin 4, not (forall y : Fin 4, PrimeOrbitClosure
x y)
```

Here `FullOrbit` is the full-orbit reachability relation (its totality is minimality), `PrimeOrbitAt` is the prime-index orbit, and `PrimeOrbitClosure` is the prime-index orbit with the closure modelled by the set itself, which is correct because X is finite and discrete. The project also has `conjecture_00000000153_false_at_zero`, the same statement specialised to $x = 0$. The axiom audit (`lake env lean Check.lean`) reports only `propxt` and `Quot.sound` for every theorem, and no `sorryAx`. See `lean4/README.md` for the full statement table and scope note.

7 Reproducibility

```
$ python3 reproduce.py
... (enumerates all primes p < 1e6; checks the residue set,
... the 4-cycle, minimality, and each prime orbit)
PASS: all checks verified; conjecture 00000000153 is FALSE.
```

`reproduce.py` uses only the Python 3 standard library. It sieves all 78498 primes $p < 10^6$, verifies that no such prime is divisible by 4 and that the residue set $\{p \bmod 4\}$ is exactly $\{1, 2, 3\}$; verifies that T is a 4-cycle and that the full orbit of every x is X (minimality); and verifies that for each $x \in \{0, 1, 2, 3\}$ the prime-index orbit is exactly $X \setminus \{x\}$, so its closure is a proper subset of X . It prints `PASS` and exits 0 exactly when every check holds, and exits non-zero otherwise.

The Lean project is built and audited by

```
cd lean4
lake build
lake env lean Check.lean
```

The document is compiled by

```
tectonic main.tex
(equivalently: tectonic --outdir build main.tex)
```

producing `build/main.pdf` (with `tectonic main.tex` the file is written to `main.pdf` in the submission root and then placed at `build/main.pdf`, the same location as the template submission).

8 Numerical vantage point

x	full orbit $\{T^n x\}$	prime orbit $P(x)$	omitted	$\overline{P(x)} = X?$
0	$\{0, 1, 2, 3\}$	$\{1, 2, 3\}$	0	no
1	$\{0, 1, 2, 3\}$	$\{0, 2, 3\}$	1	no
2	$\{0, 1, 2, 3\}$	$\{0, 1, 3\}$	2	no
3	$\{0, 1, 2, 3\}$	$\{0, 1, 2\}$	3	no

Every full orbit is X (the system is minimal), yet every prime-index orbit omits exactly one point (its base point), so no prime-index orbit closure equals X . The conjecture fails at all four points of the minimal system.

9 Status against the submission rules

- **LaTeX source** — present (`main.tex`), a standalone article using only `geometry`, `amsmath`, `amssymb`, `amsthm`, `parskip`, and `booktabs`; it compiles with `tectonic main.tex`.
- **PDF document** — present at `build/main.pdf`, produced by `tectonic main.tex`.
- **Lean 4 project** — present under `lean4/`, with `lean-toolchain` (pinned to `leanprover/lean4:v4.33.1`), `lakefile.toml` (library `Main`, project `tlmc153`), `Main.lean` (core Lean only, no `sorry`), `Check.lean` (`#print axioms audit`), and `README.md`. The audit reports only `[propext, Quot.sound]` and no `sorryAx`.
- **Reproduction** — `python3 reproduce.py` passes with exit code 0 and prints `PASS`.